

incompleteness theorem, for the general logic also developed in MN. The completeness theorems shed light specifically on the structure and potential of HI. In contrast, the general incompleteness theorem offers profound insights into the nature and limits of HI, AI, and even unknown forms of intelligence that may exist on far-distant stars.

In the mid-18th century, Carl Linnaeus, the founder of taxonomy, named the human species "Homo sapiens" — "wise being." Completeness theorems such as the above provide logical proof that humanity can indeed become truly wise. Long before that, however — perhaps even predating the origins of Judaism — the doctrine of original sin taught that all humankind bears the burden of a primordial transgression: the first humans disobeyed God and ate the fruit of the tree of knowledge. Incompleteness theorems such as the above not only offer a logical demonstration of the inherent limits of human intelligence, but also suggest a possible answer to the question: "What is original sin?" It is this: by eating the fruit of knowledge, humanity acquired a partial and incomplete wisdom. That, I propose, is the true nature of original sin. And interpreted in this way, the ancient doctrine speaks with renewed relevance in the twenty-first century. For today, humanity possesses the knowledge to exploit the natural world in myriad ways, yet lacks the wisdom to restrain that exploitation. As a result, we now suffer punishment in the form of global heating that makes life increasingly difficult and threatens the survival of our own species.

A.1. Noological Set Theory

A clarification of my view on sets is in order, prompted by the following statement in §1.2 (**Answer to Question 2**):

In the mathematical science mentioned above, the noo-worlds are defined as sets, whereas the noo-cosmos cannot be defined as a set due to a certain set-theoretical reason to be clarified in §A.1 (Noological Set Theory).

The set theory presented here is neither naïve nor axiomatic, but rather noological.

The Noological Definition of Sets

A set is defined as a collection of *entities* — called the elements of the set — within a *perceived* range. These entities are gathered by our *intellect*, and the set itself is regarded as an entity.

This definition is noological because of the italicized words "intellect," "perceived," and "entities," — an noo-element is called an entity if it makes sense to ask whether it exists.

Now, we can proceed as in naïve set theory. However, there exist certain differences. Notably, the following holds:

Proposition on Belonginess

Every set is distinct from its elements, that is, it does not belong to itself.

This is because each set is constructed by our intellect, and we cannot include the set among its elements prior to constructing it. This is analogous to the fact that we cannot include a dish among its ingredients prior to making it.

Entities other than sets are called prime entities. For example, the symbol I is a prime entity, as are all of its sequences such as I, II, III , and so on. We call the prime entities $I, II, III, \dots$ the natural numbers as the ancient romans probably did, and as usual, denote them respectively by $1, 2, 3, \dots$ in the decimal system. Using them, we can construct the sets of integers, rational numbers, real numbers, and so on, in a well-known way. The numbers are all prime entities. Thus, our set theory is open in that it takes prime entities into consideration.

Proposition on Entitic Conditions (conditions on entities)

If an entitic condition C satisfies the following condition 1, then it must also satisfy the following condition 2:

1. There exists the set of all entities satisfying C .
2. For some set X , every set containing X does not satisfy C .

However, the converse is not always true, that is, there exists an entitic condition C which satisfies condition 2 but does not satisfy condition 1.

Proof of the first assertion

Suppose that an entitic condition C satisfies condition 1. Then there exists the set A of all entities satisfying C . By way of contradiction, assume that C does not satisfy condition 2. Then there exists a set B which contains A and satisfies C . The set B is an entity satisfying C , and so is an element of A . Since A is contained in B , B is an element B , contrary to **Proposition on Belonginess**.

Proof of the second assertion

By way of contradiction, assume that every entitic condition satisfying condition 2 also satisfies condition 1. Let p be a prime entity. Define the entitic condition C_p so that an entity x satisfies it if and only if $x = p$. Then every set does not satisfy C_p , and so C_p satisfies condition 2. Therefore, C_p satisfies condition 1 by the above assumption, that is, there exists the set $\{p\}$ consisting only of p . Define the entitic condition C so that an entity x satisfies it if and only if x is a set whose elements are not prime entities. Then every set containing $\{p\}$ does not satisfy C , and so C satisfies condition 2. Therefore, C satisfies condition 1 by the above assumption, that is, there exists the set A of all sets whose elements are not prime entities. But then, A is a set whose elements are not prime entities, and so A belongs to A , contrary to Proposition on Belonginess.

An entitic condition is said to be unbounded if it does not satisfy condition 2 in Proposition on Entitic Conditions. Namely, an entitic condition C is unbounded if, for each set X , there exists a set which contains X and satisfies C . Proposition on Entitic Conditions, then, shows that there does not exist the set of all entities satisfying C . This has various consequences about mathematics:

1. There does not exist the set of all entities. This is because the empty entitic condition is obviously unbounded. Due to this, we cannot define noo-cosmos as a set, as was stated in §1.2 (**Answer to Question 2**).
2. There does not exist the set of all sets. To see this, define the entitic condition C so that an entity x satisfies it if and only if x is a set. Then C is obviously unbounded. Due to this and **Proposition on Belonginess**, we can forget about the so-called Russel's paradox in naïve set theory. Also, we cannot develop the theory of cardinalities of sets as in naïve set theory, and similarly for order types.

3. There does not exist the set of all well-ordered sets, and similarly for totally ordered sets and ordered sets. This is because every set is well-ordered by Zermelo's theorem.
4. There does not exist the set of all groups, and similarly for monoids and semigroups. This is because, for an arbitrary given set X , there exists the free group whose basis is equal to X .
5. For an arbitrarily given field K , there does not exist the set of all K -algebras, and similarly for K -vector spaces and rings. This is because every group G gives rise to a K -group algebra whose basis is equal to G .

A.2. Mistakes in Paradoxes

Since I mentioned Russel's paradox, some remarks on paradoxes may be in order. An event A is said to be paradoxical if both A and its negation $\neg A$ are necessary, that is, if $(\) \Rightarrow A$ and $(\) \Rightarrow \neg A$. In fact, no paradoxical event exists. This is verified by using the following logical laws:

1. If $A \Rightarrow B$ and $B \Rightarrow C$ for events A , B , and C , then $A \Rightarrow C$.
2. If $\neg A \Rightarrow B$ for events A and B , then $(\) \Rightarrow A \text{ OR } B$.
3. If $(\) \Rightarrow A \text{ OR } \neg B$ for events A and B , then $B \Rightarrow A$.
4. If $(\) \Rightarrow A \text{ OR } A$ for an event A , then $(\) \Rightarrow A$.
5. $\neg(\neg A) \Rightarrow A$ for each event A .

It is a good logical exercise to derive the following laws from the above ones:

6. If $\neg A \Rightarrow \neg B$ for events A and B , then $B \Rightarrow A$.
7. If $\neg A \Rightarrow A$, then $(\) \Rightarrow A$.
8. If $A \Rightarrow \neg A$, then $(\) \Rightarrow \neg A$.

Law 6 was used in the proof of **Proposition on Entitic Conditions**. The above-mentioned fact may be stated and proved as follows:

Proposition on Paradoxes

No paradoxical event exists.

Proof

Suppose A is a paradoxical event. Then $(\) \Rightarrow A$ and $(\) \Rightarrow \neg A$, and so $X \Rightarrow A$ and $\neg Y \Rightarrow \neg A$ for arbitrary events X and Y ; the latter yields $A \Rightarrow Y$ by **Law 6**, hence $X \Rightarrow Y$ by **Law 1**. However, for instance, " x is x " $\Rightarrow$ " $\neg(x$ is x)" does not hold for every entity x .

I define a paradox as a reasoning which concludes that there exists a paradoxical event, which is incorrect according to Proposition on Paradoxes. Incidentally, the prefix "para-" in particular means "incorrect," and "dox" derives from "dokein," meaning "to think."

Examples of Paradoxes and Their Mistakes

Example 1. Achilles does not catch up with the tortoise:

Achilles runs a race with a tortoise. The tortoise starts a meters ahead of him.

Achilles and the tortoise run at b meters per second and c meters per second, respectively, with b greater than c . Then, the distance between them decreases at a rate of $(b - c)$ meters per second, and becomes zero after $a/(b - c)$ seconds,

that is, Achilles catches up with the tortoise at that time. On the other hand, he reaches the tortoise's starting point after a/b seconds, by which time the tortoise has moved ac/b meters ahead. He then reaches that new point after ac/b^2 seconds, by which time the tortoise has moved ac^2/b^2 meters further. He reaches that point after ac^2/b^3 seconds, when the tortoise is ac^3/b^3 meters ahead; and so on. Therefore, the tortoise remains ahead of Achilles, always, and he does not catch up with the tortoise.

This reasoning concludes that both the event "Achilles catches up with the tortoise" and its negation are necessary; hence, it is paradoxical. According to **Proposition on Paradoxes**, there must be a mistake here. The total time during which the tortoise remains ahead of Achilles is less than the infinite sum $(a/b + ac/b^2 + ac^2/b^3 + \dots)$, which is equal to $a/(b-c)$ seconds — the exact time at which Achilles catches up with the tortoise. Therefore, the description following "On the other hand" up to "and so on" at most means "The tortoise remains ahead of Achilles, until he catches up with it," which is correct. The mistake lies in replacing "until he catches up with it" with "always."

Example 2. There is a declarative which is both false and not false:

Consider the declarative "This declarative is false." If this declarative is false, then it is not false. If it is not false, then it is false.

Let A denote the event "This declarative is not false." Then this reasoning concludes that $\neg A \Rightarrow A$ and $A \Rightarrow \neg A$. Hence it is paradoxical by the **laws 7 and 8** above. According to **Proposition on Paradoxes**, there must be a mistake here. Let B denote the first underlined "this declarative." Then the second underlined "this declarative" refers to the declarative "B is false." Let C denote it. Then the above reasoning in fact concludes that B is false if and only if C is not false, which is correct. Hence B is not equal to C. Probably, misled by the ambiguity of the word "this declarative," the above reasoning assumed that B was equal to C, hence concluded that B and C were both false and not false.

Example 3. Inexpressible entities are expressible:

Let a denote one of the entities which are inexpressible. Then a is expressible as "one of the entities which are inexpressible," and so a is not one of the entities which are inexpressible. Thus, a is not a .

This reasoning concludes that the event " a is not a " is necessary, hence paradoxical, because its negation " a is a " is also necessary. According to **Proposition on Paradoxes**, there must be a mistake here. In fact, after expressing an entity as "an inexpressible entity," it is lying as "it is an inexpressible entity." There is another mistake. Every entity is expressible, even by one letter, as has been shown by the above sentence "Let a denote one of the entities which are inexpressible." Similar remarks apply to the paradoxes obtained by replacing "express" by "explain" and "define."

If x is an entity and A is a set, the expression " $x \in A$ " means " x belongs to A ." Russel's paradox in naïve set theory is as follows:

Let A be the set of all sets X which do not satisfy $X \in X$. Then, if A satisfies $A \in A$, then A does not satisfy $A \in A$ by the definition of A . If A does not satisfy $A \in A$, then A satisfies $A \in A$ by the definition of A .

Let B denote the event " A does not satisfy $A \in A$." Then Russel concluded above that $\neg B \Rightarrow B$ and $B \Rightarrow \neg B$. Hence B is paradoxical by the **laws 7 and 8** above. According to **Proposition on Paradoxes**, there must be a mistake here. As shown above, it was found out by proceeding to noological set theory, which is open and broad. It is a pity that the initiators of naïve set theory could not find out it, and some of them proceeded to axiomatic set theory, which is closed and narrow.

A.3. Mathematical Science and Models

(to be written)

2. Life and the Biosphere

As stated in the **Introduction**, the starting point of Eusophy is **Declaration (6)**: "Humankind is but one among many species." Therefore, a consideration of life is indispensable to this treatise.

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1. **Systems with Life and Survival Instinct**
2. **Various Aspects of Life**
3. **Species**
4. **Evolution**
5. **The Biosphere for Humankind**

2.1. Systems with Life and Survival Instinct

By definition, a life system is a system whose purpose is its own survival — that is, the continuation of its life — and whose life is the process of autonomous self-reproduction of its components, including itself as a whole. Purposes are inseparably linked to orientations: every purpose necessarily gives rise to an orientation toward itself, and conversely, every orientation presupposes a corresponding purpose. Therefore, a life system can also be defined as a survival-oriented system. Its survival instinct is thus defined as its orientation toward survival.

Terrestrial life systems — animals, plants, bacteria, and so on — are conventionally referred to as organisms. Even viruses qualify as life systems in this framework. However, they can be life systems only within other life systems, just as naked humans can be life systems only within the environment of the Earth. Extraterrestrial life systems, as conceived here, may not require organic matter — that is, carbon-based compounds — for their structure or survival.

According to this definition, every life system engages in a struggle for the continuation of its life through its survival instinct. The survival instinct of bacteria and viruses causes many human diseases and gives rise to many antibiotics, some of which are beneficial to humans.

The survival instinct of plants can be observed through time-lapse photography. Notable examples include the following YouTube videos:

- **How Plants Climb — Time Lapse**
- **Changing the Direction of the Light During Plant Growth — Time Lapse**
- **These Seeds Can Walk! | The Green Planet | BBC Earth**
- **The Tyrant of the Deep | The Green Planet | BBC Earth**
- **Hungry Venus Flytraps Snap Shut on a Host of Unfortunate Flies | Life — BBC**

Struggle for life is not necessarily competition with other life systems. Rather, it is essentially a strenuous effort in the face of difficulty in continuing life — specifically, in reproducing its components, including itself. As will be shown in §2.4 (**Evolution**), competition is not the central concept in evolutionary theory, contrary to widespread belief. As suggested by **Observation (8)** and to be discussed in §2.5 (**The Biosphere for Humankind**), the biosphere is a system sustained not only by competition but also by cooperation.

2.2. Various Aspects of Life

A life system, being a system, necessarily possesses a clear boundary between its interior and exterior. Since its purpose is the continuation of its life — that is, the persistence of its autonomous self-reproduction process — it must maintain within itself a blueprint and a procedural scheme for self-reproduction. It must also, as needed, acquire materials and sources of energy from its surroundings, store them internally, and discharge surplus substances back into the environment.

These acts of intake, storage, and discharge are all subject, to varying degrees, to the conditions of both the external and internal environments. Consequently, even though the self-reproduction of a life system is autonomous and directed toward its own continuation, it is neither entirely free nor constant. It is constrained to some extent by those conditions, and when the constraints are severe, the process may cease altogether. When the constraints are mild, however, the process can continue, adapting to the conditions as necessary. It should be noted that the external environment of one life system may include the self-reproduction activities of other life systems.

The replicas produced by a life system — whether of itself or of its components — may emerge either within or outside the system. Even if initially produced internally, they may eventually be released into the external environment.

Moreover, these replicas are not necessarily exact copies; they may vary in various ways. Even when they are accurate copies, they may undergo changes after production.

Moreover, such replicas may either be produced as life systems themselves or become life systems after their production. In either case, the original life system is called the parent, and the resulting life system is called the offspring. The offspring of the offspring, and so on, are called descendants.

Strictly speaking, an offspring is distinct from its parent, but it may also be regarded as continuous with the parent. If so regarded, each individual life continues until its descendants come to an end. This continuity may be likened to a factory that, even after being rebuilt

multiple times, is still considered the same factory as long as the same production process continues. This remark leads to my view of life and death, to be presented in §3.1 (**Baton of Life**).

Since self-reproduction is constrained by internal and external conditions, it is more readily achieved in environments where such constraints are minimal. Moreover, as noted above, self-reproduction inevitably involves variation. Depending on the attributes resulting from such variation, the degree of constraint imposed by the environment may change. Thus, as the production of offspring continues — offspring of offspring, and so on — groups of life systems possessing particular attributes may gradually increase or decrease in number depending on their compatibility with internal and external conditions. This principle may be called the principle of gradual change, as it gradually gives rise to various phenomena, including those to be discussed in the succeeding paragraphs.

As stated in §2.1 (**Systems with Life and Survival Instinct**), a life system is a system endowed with a purpose of survival. From this purpose arises the effort for survival — that is, the effort to achieve the purpose of survival. In some life systems, this effort gives rise to intelligence, which serves the attainment of survival. Depending on the degree to which the purpose of survival is fulfilled, emotion may also arise. Moreover, will emerges as a drive toward achieving survival, and the capacity for memory — that is, for retaining and recalling knowledge acquired through intelligence — also develops. In this way, the mental faculties of intelligence, emotion, volition, and memory gradually arise from the structure of the life system oriented toward survival. The culmination of such faculties may be found in the Self (SS) as defined in §1.8 (**The Self, Values, Emotion, Volition, and Memory**).

Moreover, the emotions that correspond to the degree of survival fulfillment may give rise to purposes beyond mere survival, and to efforts beyond mere survival efforts. These new purposes and efforts, in turn, give rise to evaluations of whether they are compatible or incompatible with survival. When the living environment is insufficient relative to the number of life systems, some of these efforts — including efforts for survival — may transform into competition among life systems, particularly into competition for survival, that is, competition aimed at achieving the purpose of survival.

As stated in §1.2 (**Answer to Question 2**), the noo-cosmos is the totality of noo-elements, and intelligence is the capacity to know the noo-cosmos. The noo-elements are divided into entities and events, and each point in time or location in space is regarded as an entity. Therefore, even in life systems other than humans, if they possess sufficient intelligence, a form of self-consciousness may gradually arise — namely, the perception of the boundary of the life system within space-time. This refers to the perception of the surface of a life system at the present moment — the boundary between the internal and external aspects of the life system, at the boundary between the past and the future. Thus, in this treatise, we define:

Self-consciousness = Perception of the boundary of the life system in space-time.

However, since the term "self-consciousness" is polysemous, we shall refer to this specific sense as self-boundary perception from here onward.

In **MP**, I discussed three types of self-consciousness: degree of wakefulness, cognition of one's own thoughts, and sense of self-existence. In this context, cognition corresponds to what this treatise calls perception, and the sense of self-existence may be understood as arising from the integration of self-boundary perception, wakefulness, cognition of one's own thoughts, and memory.

If we study human self-boundary perception as a form of self-consciousness in mathematical science as discussed in §A.3 (**Mathematical Science and Models**) — though it is beyond our present scope — the object of consciousness may be regarded as certain extremely high-dimensional vector-valued function with time as the independent variable. The elements of the vector represent stimuli to sensory cells at each moment. Consciousness itself may then be defined as the derivative or difference of that function. Thus, the process of consciousness can be understood as a process of differentiation or differencing within the nervous system. This mode of understanding is akin to the microscopic approach of neuroscience, which stands in contrast to the macroscopic approach of **MN** as explained in §1.7 (**Intellection, Perception, and Ratiocination**). Because of this contrast in perspective, neither **MN** nor **MP** treats self-consciousness as a primary object of study.

2.3. Species

Answering the question "What is a species?" appears to be a difficult task in taxonomy, especially when it comes to classifying extinct organisms known only from fossils. However, Eusophy is oriented toward the future — specifically, toward the survival of humankind as a species. Accordingly, in this treatise, when I refer to humankind or living beings, I generally mean future humankind or future living beings. Since the present and even the recent past were, not long ago, part of the future, the term "future" effectively includes the present and the near past as well.

When it comes to future living beings, the question "What is a species?" is not so difficult to answer. I define two individuals as belonging to the same species if they are capable of continuous interbreeding — that is, if they can mate and produce offspring that can, in turn, mate with the original individuals. This is not merely a definition; if it becomes necessary to determine whether two individuals belong to the same species, the determination should be made according to this definition.

Such determination does not require direct mating between the individuals in question or between parents and offspring. It suffices to test mating compatibility with preserved specimens. For this purpose, if feasible, we should freeze and store multiple samples of eggs and sperm from various organisms, updating the specimens periodically.

To illustrate: although humanity comprises many races, the history of intermarriage among them has demonstrated that humans are capable of continuous interbreeding. This confirms **Declaration (6)**: "Humankind is but one among many species." In fact, we must define the scope of humankind itself. I propose that any individual capable of continuous interbreeding with human specimens be defined as belonging to humankind. This definition alone suffices; it is not necessary to determine, for every individual, whether it belongs to humankind.

Horses and donkeys can interbreed, but their offspring are sterile and thus not capable of continuous interbreeding. Therefore, they are distinct species. In contrast, pigs and wild boars are likely the same species, since pigs are domesticated boars. Similarly, dogs and wolves may also be the same species, given their shared ancestry and interbreeding potential.

To divide organisms into species based on continuous interbreeding, the transitivity law must hold: if individual A can continuously interbreed with individual B, and B with individual C, then A must also be able to continuously interbreed with C. This law is both necessary and sufficient for such a classification.

If the transitivity law fails, or if continuous interbreeding cannot be determined in principle, or if the organisms in question do not reproduce through mating at all, then — if classification remains important — it should be carried out using alternative criteria appropriate to the reasons for its importance. The same applies to organisms such as cultivated plants and domesticated animals, which have undergone extensive artificial breeding and for which classification by continuous interbreeding may not be appropriate.

There is little need to be overly concerned about how to classify species, because the only taxonomically important fact in this treatise is the above-mentioned Declaration (6), and the manner in which species are classified does not affect the explanation of evolution in the next subsection.

2.4. Evolution

In this section, I do not aim to define the standard meanings of the major terms in evolutionary theory, but rather to present my own interpretation. Let us begin by stating what I shall call the principle of evolution — also referred to as the principle of gradual change, as mentioned in §2.2 (**Various Aspects of Life**):

Within a single species sharing a common habitat, if individuals possessing a certain genetic trait leave more offspring than those lacking it, then the proportion of individuals with that trait will, over successive generations, approach 100%.

The validity of this principle can be grasped using only elementary arithmetic. Suppose we make a count of the individuals with that trait at fixed intervals. Then we obtain a series $a_0, \dots, a_{n-1}, a_n, \dots$ of numbers. Here, the subscript 0 represents the time of initial count, and 1 in the subscript $n-1$ represents the interval, such as one year, five years, ten years, and so on. Let $b_0, \dots, b_{n-1}, b_n, \dots$ be the series of numbers similarly obtained for the individuals lacking that trait. Then the proportion in question is equal to $100a_n/(a_n + b_n)$, which is equal to $100/(1 + b_n/a_n)$. For $n = 1, 2, \dots$, let r_n and s_n denote the ratios a_n/a_{n-1} and b_n/b_{n-1} , respectively. Then $a_n = r_n r_{n-1} \cdot \dots \cdot r_1 a_0$ and $b_n = s_n s_{n-1} \cdot \dots \cdot s_1 b_0$, and so $b_n/a_n = (s_n/r_n)(s_{n-1}/r_{n-1}) \cdot \dots \cdot (s_1/r_1)(b_0/a_0)$. Now, the assumption of the principle of evolution implies that $s_n < r_n$, that is, $s_n/r_n < 1$ for each n . Here, assume more strongly that there exists a number q such that $s_n/r_n < q < 1$ for each n . Then $b_n/a_n < q^n(b_0/a_0)$ for each n . As n continues to increase, b_n/a_n converges to 0 because $0 < q < 1$, and so $100/(1 + b_n/a_n)$ converges to 100, as asserted in that principle.

The conclusion of the principle of evolution — that the proportion of individuals possessing a certain genetic trait approaches 100% over successive generations — does not necessarily mean that the proportion will reach 100%. As the proportion of individuals with the trait increases, the habitat may change — for example, through pollution, depletion of food sources, or loss of suitable living space. In such cases, individuals with the trait may no longer leave significantly more offspring than those without it. As a result, the proportion of individuals with the trait may stabilize at, say, 70%, or even begin to decline.

Assume as in the principle of evolution that individuals possessing a certain genetic trait leave more offspring than those lacking it. Then we may say that the trait — or the individuals possessing it — is adapted to the habitat, or that it represents the fittest in that environment. If the habitat is subject to human influence, as in the case of cultivated fields or breeding facilities — or even if not — we may personify the environment and say that the trait or individuals have been selected by the environment. We may also describe the outcome of the principle by saying that the trait or the individuals possessing it tend toward plenitude — that is, toward being fully present throughout the habitat. Thus, the principle may be restated in various ways:

- Within a single species sharing a common habitat, genetic traits or individuals adapted to that environment tend toward plenitude.
- Within a single species sharing a common habitat, the fittest tend toward plenitude.
- Within a single species sharing a common habitat, genetic traits or individuals selected by the environment tend toward plenitude.

In the Book of Genesis in the Old Testament, there is a well-known story of Noah's Ark. According to the narrative, when evil had spread throughout human society, God regretted having created humankind. He instructed Noah, a righteous man, to build an ark, and then sent a great flood to destroy all humans except Noah and his family. After the flood, God once again spoke the words He had originally addressed to the first humans: "Be fruitful, and multiply, and replenish the earth."

Strikingly, this episode can be interpreted as a parable that illustrates the principle of evolution described above. That is, the individuals (Noah and his descendants) who were selected by the environment (represented by God) go on to tend toward plenitude — they are fruitful, they multiply, and they fill the earth. This interpretation is by no means far-fetched. The Book of Genesis expresses the worldview and understanding of life held by ancient peoples, shaped in a time when civilization — understood, as in **Observation (7)**, as the totality of human knowledge — was still in its infancy. If we read such texts as parables and seek their underlying meanings, we may find that they resonate with modern science and can even be harmonized with a scientific worldview and understanding of life.

Let us suppose that, by some principle distinct from the principle of evolution, a single species of life first emerged in the primordial world. On that basis, I propose the following as the law of evolution, and assume it to be valid:

Through variation during autonomous self-reproduction, plenitude, migration, and repeated dispersion and aggregation, various species have emerged, undergone changes in their genetic traits, and either survived (as species) or failed to survive.

Moreover, I shall refer to the changes in genetic traits of a species — caused by variation during autonomous self-reproduction, plenitude, migration, and repeated dispersion and aggregation — as the evolution of that species or of those traits. In this sense, "evolution" refers to changes that occur over successive generations, and in most species, such changes are gradual.

The global spread of the novel coronavirus that began in 2020 provides empirical support for the validity of this law of evolution. The virus that causes COVID-19 undergoes various mutations as it replicates within the human body. Through transmission from person to person, it migrates and undergoes repeated dispersion and aggregation. Variants with higher infectivity — those that leave more offspring — are the fittest within the habitat of the human body and tend toward plenitude. In this way, they survive.

As stated in §2.1 (**Systems with Life and Survival Instinct**), viruses are life systems within the human body. That the law of evolution applies to such viruses is one piece of empirical evidence supporting its validity.

2.5. The Biosphere for Humankind

The aim **(2)** of Eusophy — to explore the path to the survival of humankind — does not imply that the survival of other species is to be disregarded. Of course, there are species such as *Mycobacterium tuberculosis* whose survival we may justifiably seek to eliminate. However, the survival of humankind depends on the survival of many other species. The reasons for this are somewhat intricate.

Let us begin by imagining a time before the advent of animal husbandry, agriculture, or aquaculture, when humans obtained food through hunting, gathering, and fishing. In those times, humans survived by consuming various wild organisms, each of which, in turn, had survived by consuming other organisms. In this way, humans lived within a food chain — a chain of eating and being eaten. This food chain is itself long, but it also connects to even longer chains. The remains of organisms consumed and excreted by animals are further consumed and excreted by other animals, or ultimately decomposed by microorganisms into soil components, or transformed into atmospheric CO₂ (carbon dioxide) through combustion or slow oxidation. Over long periods, these remains may also become carbon compounds in the earth or sea, eventually returning to the atmosphere as CO₂.

Yet even this is not the end. The remains that have become soil components or atmospheric CO₂ are absorbed by plants — through their roots or via photosynthesis — and become part of plant bodies. These plants are then consumed by animals, and the cycle continues. This broader network of relationships, which includes but extends beyond the food chain, forms a system in which "consuming and being consumed" occurs in a wide sense, and in which matter circulates and returns to its origin. Rather than the commonly used term "material cycle," I shall refer to this as the food cycle.

Even today, when most of our food comes from animal husbandry, agriculture, and aquaculture, the food cycle continues. While hunting and gathering have diminished, fishing has expanded. All living organisms, including humans, continue to survive within the food

cycle. In **Observation (8)**, I provisionally defined the food cycle as the composite of the food chain and the carbon cycle, where the latter refers to the circulation of carbon among the atmosphere, the earth's crust, the oceans, and living organisms through various carbon compounds.

The relationships among organisms that support survival are not limited to the food cycle. Many organisms survive through symbiosis — mutual support among different species. It is now well known that animals live in symbiosis with bacteria in their bodies. Plants also engage in symbiosis — with fungi, with other plants, and with soil organisms through networks of roots and mycelia. When insects or birds feed on nectar or pollen and assist in pollination, this too is a form of symbiosis. Insects that damage plants may be eaten by birds, or seeds consumed by birds may be dispersed and deposited elsewhere, allowing plants to expand their habitat — these are also forms of symbiosis. Even human cultivation of plants and domestication of animals can be seen as symbiosis: those plants and animals survive by being utilized by humans. Biologists believe that organelles such as mitochondria and chloroplasts, which support the survival of many organisms, originated from bacteria that once lived symbiotically within primitive cells.

These are examples of interspecies symbiosis, but intraspecies symbiosis also exists. Social insects such as ants and bees, and other social organisms that live in structured communities, can be seen as engaging in intraspecies symbiosis. So too can sexually reproducing organisms, in which males and females of the same species coexist and cooperate. In fact, multicellular organisms themselves may be viewed as symbiotic collectives of single-celled organisms of the same kind. Their reproductive cells correspond to the females or queens in social species. Or rather, we may imagine that primitive multicellular organisms, in seeking greater efficiency in survival, ceased expanding as single bodies and instead evolved into social or sexually reproducing organisms. The size and structural complexity of such organisms may have evolved to the extent that maximizes survival efficiency.

The relationships among organisms that support survival are not limited to the food cycle and symbiosis. In many cases, the components of an organism's habitat are provided by other organisms. I shall refer to this relationship as habitat bioprovision. For example, forests, grasslands, seagrass beds, and coral reefs — key components of many animals' habitats — are created by plants or other animals. The soil that supports plant life is formed from organic remains by insects, fungi, and microorganisms. In parasitism or infection, the body of one organism becomes the habitat of another, often without mutual benefit — unlike symbiosis. What, then, constitutes the habitat of humankind? Even a person living in a high-rise apartment in a concrete jungle depends on food, clothing, and shelter made from materials produced around the world. Their habitat thus includes the habitats of those who produce such materials — farmlands, pastures, fisheries, forests — all of which are shaped and sustained by living organisms. The oxygen that sustains nearly all life today was first produced by cyanobacteria in ancient times and continues to be generated by plants.

From these examples alone, it is clear that the survival of any species depends on the survival of many other species and the sustainability of their habitats, through the food cycle, symbiosis, and habitat bioprovision.

By definition, the biosphere is the system composed of all biological communities and their habitats. I shall refer to the network of connections described above as the biospherical network: the network of food cycles, symbioses, and habitat bioprovisions that link various biological communities and their habitats within the biosphere.

Thus, the biosphere is a system in which biological communities and their habitats are interconnected through the biospherical network. As the title "Life and the Biosphere" of this section suggests, the survival of any species depends on the survival of many other species that are linked within the biosphere and on the sustainability of their habitats. And as stated in **Declaration (6)**, humankind is merely one among many species and part of the biosphere. Therefore, I go one step further than the earlier claim that "the survival of humankind depends on the survival of many other species," and assert the following:

The survival of humankind depends on the sustainability of the biosphere.

The biosphere, however, is also a dynamic system. It has reached its current state through a long process of transformation, and this process will likely continue. What matters is that this transformation proceeds in a direction favorable to humankind — that is, that the biosphere does not evolve into a state that makes human survival more difficult. Ideally, this favorable transformation should continue for as long as possible — perhaps for billions of years, as long as solar energy remains available. As noted earlier, there may be species such as *Mycobacterium tuberculosis* whose extinction is acceptable. Therefore, the "sustainability of the biosphere" mentioned above does not mean preserving the biosphere in its current form. Rather, it means ensuring that its ongoing transformation continues in a direction that is as favorable as possible to human survival. This is what I meant by the title "The Biosphere for Humankind" of this subsection.

3. View of Life, Death, and Happiness

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3.1. Baton of Life

People die in many different ways. If we were to regard all forms of death as misfortunes, then — since everyone must eventually die — it would follow that every human being, including the newborn child whose birth is now being celebrated, is destined to end in misfortune. I do not wish to hold such a view of life and death.

Biology suggests a view of life and death that stands apart — not only from the notion that all death is misfortune, but also from religious and ancient views. The reproductive cells produced by your body, when united with those of another, can undergo repeated cell division and develop into a new human being. While the other cells in your body will eventually age and die, your reproductive cells may live on as a new person who inherits your genetic traits. If health permits and no other events intervene, this process can continue unbroken from

generation to generation. In this sense, every human being is originally endowed with eternal life.

To put it metaphorically, we may think of life as batons. Each individual life is like a baton in a relay race — one without a finish line, without winners or losers. Each of us is entrusted with the vital role of receiving the baton and passing it on, as a member of a team. If one can leave the stage with the peace of mind that one has fulfilled this role, then such a departure can hardly be called unfortunate. Moreover, even after fulfilling this role, one does not usually exit immediately; rather, one often remains to witness the baton being passed on, or even to assist in the passing. This, too, may be seen as a reward for having fulfilled one's role. Or perhaps it is a mistaken sense of completion, and in truth, we are often being urged to pass on many more batons.

These urgings, these rewards, and the very role of passing on the baton of life — all of them are granted to us by the principle of life described in §2.1 (**Systems with Life and Survival Instinct**). It is this principle that allows even the most innocent of girls to grow into a child, then a maiden, fall in love, become a mother, a grandmother, and eventually an ancestor with countless descendants. The same is true for boys. In this way, the baton of life continues to be passed on.

In our modern, scientifically advanced age, I believe that spreading this biologically grounded view of life and death is a sound and compassionate way to help people overcome fear and sorrow in the face of death — both their own and that of others. Moreover, such a view brings with it a beautiful byproduct: a deepening sense of affection and compassion for one's parents and children, and by extension, for one's ancestors and descendants. For your ancestors and descendants are, in a profound sense, your past and future selves. Love for one's descendants naturally leads to the desire to contribute to the survival of humankind — for it is only through the continued existence of humanity that your descendants can live.

This desire lies at the very foundation of Eusophy proposed in **FE** and developed throughout this treatise, as well as of the answer to **Universal Question (1)** "What should anyone do?" Such a view of life and death may also inspire children to embrace the goal of becoming able to pass on the baton of life. And for children who suffer so deeply — perhaps due to bullying at school — that they feel driven to end their lives, this view may offer a reason to live: to stay alive until they can pass on the baton they have received. For to pass on the baton of life is to meet one's future self in the next generation — a thrilling and hopeful prospect.

Incidentally, in Mozart's opera *Die Zauberflöte* (The Magic Flute), there is a joyful duet, Pa-Pa-Papageno, in which a loving couple sings of their future together and the many children they hope to have. Just before this scene, the man had been on the verge of taking his own life. This has led to the popularization of the Papageno effect as a means of suicide prevention in adults. Yet this song may also have a powerful message for children: it tells them that their parents are overjoyed to have become their parents, and it may awaken in them the aspiration to become someone who can pass on the baton of life. In this way, it may help children find happiness and hope.

3.2. Human Flourishing

Declaration (6) and **Observation (7)** show that humankind is a species distinguished from other species by its intellect and its civilization — the totality of human knowledge acquired through intellect. §3.1 (**Baton of Life**) offers a view of the meaning of life. Taken together, these point toward what may be called human flourishing — a form of happiness realized through passing on the baton of life and knowledge: through diligent learning at home, in school, and at work; through forming a family and welcoming children into one's family; and through raising those children while handing down the knowledge one has learned.

At the societal level, such flourishing can be assessed objectively through a constellation of indicators — including marriage and birth rates, school enrollment and advancement, employment, and the accessibility and quality of services in healthcare, childcare, education, and public communication.

4. Causality in Science, Logic, Ethics, and Religion

Abstract: Modern science, logic, ethics, and religion are each intellectual endeavors aimed at understanding causality. Scientists each pursue laws which govern aspects of causality. Logic is the knowledge of causality in terms of validity among the declaratives. Ethics is the study and teachings of deeds which are supposed to be causally connected with moral good. Religion is, in essence, a system of speculation and prayer concerning causality and the acceptance of their outcomes — a system that began in plain form and was later embellished to cultivate such acceptance among people. Causality was defined in §1.3 (**Causality and Necessity**). "Understanding causality" means reasoning — that is, the act of pursuing the reason why $A_1, \dots, A_n \Rightarrow B$ for some tuple $(A_1, \dots, A_n, B)$ of events.

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4.1. Causality in Science

Science is the totality of endeavors to discover, with evidence, the laws that govern aspects of causality

It is a well-known story: in the seventeenth century, Newton's theory of mechanics resolved seemingly unrelated questions — such as "Why does an apple fall from a tree?" and "Why do planets orbit the sun?" — by appealing to the law of universal gravitation and the laws of motion, which describe how forces acting on a body affect its motion. In fact, Newton went further. He also answered questions like "How does an apple fall from a tree?" and "How does a planet orbit the sun?" — as explained in §A.3 (**Mathematical Science and Models**).

This story can be recast, somewhat in the language of logic, as follows: Let A denote the event "X is an apple that has detached from a branch," and let B denote the event "X falls." Here, X represents an entity — as distinguished from events in §1.2 (**Answer to Question 2**). From our experience, every apple that detaches from a branch falls. That is, for any X, if A occurs, then B occurs. In other words, A is causally connected with B in our experience.

Similarly, observation shows that the event "X is a planet" is causally connected with the event "X orbits the sun." Indeed, the celestial bodies that once appeared to wander across the sky — hence the name "planet," derived from the Greek "planētēs," meaning "wanderer" — were found by Kepler and others to orbit the sun. Newton mathematically proved that these empirically observed causal connections are governed by the law of universal gravitation and the laws of motion. That is, he demonstrated that one aspect of causality — the motion of physical bodies — is governed by these laws.

Generalizing from this, we may say, as above, that science is the totality of endeavors to discover, with evidence, the laws that govern aspects of causality. This is why science is said to be universal. It is also why science is considered objective — and why I believe that scientists are among the most sincere and trustworthy individuals when it comes to expressing knowledge: they pursue evidence.

Evidence, however, is not limited to mathematical proof. It also includes empirical methods such as experimentation and observation. Even mathematical proofs rely on mathematical models — as discussed in §A.3 — which are themselves based on empirical methods. In this sense, mathematical proof is also a kind of empirical evidence. Therefore, even laws once believed to be correct may be revised when experimental or observational techniques improve. Indeed, Newton's theory of mechanics was revised 240 years later by Einstein's theory of relativity.

Some may criticize science for being subject to revision and conclude that scientists are untrustworthy. But it is precisely because scientists continually pursue the best available evidence that such revisions occur. For this reason, I maintain that scientists are among the most sincere and trustworthy individuals in expressing what they know.

Newtonian mechanics is also a prime example of the mathematical sciences described in §A.3. As I have noted on various occasions, the contents of **MP**, **MN**, and **FE** likewise belong to the domain of mathematical science. So too does Mendelian genetics, as mentioned in §1.6 (**The Cosmos, Intellect, and Language**).

Science, understood as an intellectual endeavor to discover — with evidence — laws that govern aspects of causality, is not confined to laboratories. In fact, it is something that everyone engages in, often unconsciously, in the course of daily life. For example, in relation to food, clothing, and shelter — aspects that concern basic human needs — anyone might discover, through empirical evidence, causal laws such as: "If I overeat, I feel uncomfortably full," "If I wear many layers, I stay warm," or "If I stay indoors, I feel relaxed." These are, in a sense, scientific insights.

However, this kind of science might be more appropriately called amateur science. In this treatise, unless otherwise specified, the term "science" refers to the kind conducted in

laboratories, and "scientists" are those who engage in such science. Statements such as "Modern scientific knowledge is generally accurate and advanced" do not refer to amateur science. Amateur science is often neither advanced nor accurate, and it is not uncommon for it to contain serious errors.

I once heard a story about a certain premodern society. There, people had noticed that when lice fled from a sick person's body, death often followed soon after. They came to fear that "people die because the lice leave," and so they would deliberately place lice on their bodies. Of course, the truth is the reverse: "Lice leave because the person is dying." Not only had they confused cause and effect, but they were also unaware of a more serious fact — that being bitten by lice increases the risk of infectious disease.

We should not be too quick to laugh at such societies. In our own, for instance, some people learn that "many individuals with blood type A tend to be cautious" and conclude, "So being type A makes one more cautious." Others reject this, saying, "Blood type couldn't possibly influence personality." Both positions make the same mistake: they oversimplify the relationship between cause and effect. The truth may be something like this: "People with blood type A tend to have more biological vulnerabilities, and over the course of evolution, those who were more cautious were better adapted to their environments and left more offspring." If so, this would be a significant insight — especially for those with blood type A.

It is worth noting that, just as there is amateur science, there may also be amateur scholarship. For example, the term "philosophy" in everyday language often refers not to academic philosophy, but to fragments, remnants, or faint wisps of amateur thought. In this treatise, however, amateur scholarship other than amateur science lies outside our scope. Thus, unless otherwise specified, suffixes like "ology" or "ologist" refer to academic disciplines and those who engage in them within research institutions.

4.2. Causality in Logic

Logic is the totality of endeavors to understand causality by means of the laws that govern the assumption–consequence relation among declaratives

Let us now consider, in light of the previous subsection §4.1 (**Causality in Science**), the following formulation: Let A denote the event "X is Y," B denote the event "X does Z," and C denote the event "All Y do Z."

In §4.1, Y referred to specific entities, such as "apples that have detached from a branch" or "planets," and Z referred to specific acts, such as "fall" or "orbit the sun." Accordingly, C represented specific events such as "All apples that have detached from a branch (do) fall" or "All planets (do) orbit the sun."

In standard logic textbooks, one often encounters the example where X, Y, and Z are "Socrates," "a human," and "be mortal," respectively. In that case, A, C, and B correspond to the events "Socrates is a human," "All humans are mortal," and "Socrates is mortal."

Now, we all know that — regardless of what X, Y, and Z may be — if both A and C occur, then B occurs. That is, the events A and C together are causally connected with B, as defined in

§1.3 (Causality and Necessity).

As we humans grow intellectually, we naturally acquire the ability to connect (A, C) with B, and to perform similar operations. However, in daily life, we do so without conscious awareness, and we rarely distinguish among the various forms these operations may take.

Logicians, by contrast, have identified and distinguished such operations, discovering laws like the syllogism — which asserts that, for any X, Y, and Z, the pair (A, C) is causally connected with B. Indeed, it was likely such discoveries by ancient philosophers that marked the beginning of logic as a discipline.

The syllogism is implicitly at work in §4.1 as well. What Newton did, in effect, was to prove — using mathematics and the laws of universal gravitation and motion — that C necessarily occurs for Y and Z specified as above. Combining this with the syllogism, we can conclude that, for any X, A is causally connected with B for those specific Y and Z.

In fact, there are other logical laws also referred to as syllogisms. In **MN**, the validity of the syllogism described above is explained on the basis of case logic — a framework developed therein.

As noted above, we used A to denote the event "X is Y." Strictly speaking, however, "X is Y" is the linguistic expression of event A. The same applies to the relationship between "X does Z" and B, and between "All Y do Z" and C. Thus, X and Y represent general linguistic expressions of entities, and Z represents a general linguistic expression of an act of an entity.

As explained in §1.2 (**Answer to Question 2**), linguistic expressions of entities and events are called nominals and declaratives, respectively. Linguistic expressions of acts are called predicates — since acts are a kind of event, predicates are a kind of declarative expression.

Accordingly, the syllogism is, in essence, a law concerning declaratives. It states that, for any nominals X and Y and any predicate Z, if the declaratives "X is Y" and "All Y do Z" are both true, then the declarative "X does Z" is also true.

As written in **§1.3**, when the occurrence of certain events leads to the occurrence of other events, we call the former the causes and the latter the effects. Similarly, when the truth of certain declaratives leads to the truth of other declaratives, we may call the former the assumptions and the latter the consequences.

Thus, for any nominals X and Y and any predicate Z, the syllogism connects the pair of declaratives ("X is Y," "All Y do Z") with the declarative "X does Z" in the assumption–consequence relation. This corresponds to the causal connection between the pair of events (A, C) and the event B.

Generalizing from this, we may say that modern logic is the study of the laws that govern the assumption–consequence relation among declaratives. To discover such laws is to discover the laws that govern causality across the diverse universes — the noo-worlds — that constitute the multiverse of the noo-cosmos.

However, in this treatise, I use the term "logic" to refer more broadly to any endeavor to understand causality by means of the laws that govern the assumption–consequence relation among declaratives. The adjective "logical" is derived from this broader sense of "logic."

Taken together with the previous subsection §4.1, which defined science as "the totality of endeavors to discover, with evidence, the laws that govern aspects of causality," we can see that modern science and logic share a common feature: they both aim to understand causality through evidential laws.

This view of logic is well suited to the concerns and methods of AI engineering, which asks: "How can we enable machines to judge — or be made to judge — causality mechanically?" Of course, if the concerns or methods differ, other perspectives may be more appropriate. For example, in the noology of MN, logic is used in order to construct and analyze a mathematical model of the noo-trinity described in §1.7 (**Intellection, Perception, and Ratiocination**).

4.3. Causality in Ethics

Ethics is the speculative study and teaching of actions that are causally connected with being morally good

Confucianism is often regarded as a religion, but it is more accurately understood as a form of ethics. One of its principal texts, the "Analects," is a record of the sayings and actions of Confucius, the founder of Confucianism.

Analects 14:25 (Xian Wen 25) is a brief passage: "The Master said: In ancient times, those who studied did so for themselves; today, they study for others." Setting aside the opening phrase "The Master said," the standard interpretation is that this is a teaching of social morality: "If you wish to be morally good, study not to gain recognition from others, but to cultivate yourself." Here, "those who study" likely refers to all who engage in learning, including students. A similar interpretation would be: "If you wish to be morally good, study not to surpass others, but to improve yourself." This is because, as the examples below suggest, being recognized by others often amounts to defeating one's competitors.

Analects 2:3 (Wei Zheng 3) is another brief passage: "The Master said: Guide the people with laws and regulate them with punishments, and they will avoid wrongdoing but lack a sense of shame. Guide them with virtue and regulate them with ritual, and they will develop a sense of shame and become upright." This is a teaching of political morality: "If you wish to promote moral conduct, guide the people not with laws and punishments, but with personal virtue and propriety."

Generalizing from these examples, Confucianism may be described as a system of teachings on morally appropriate conduct in society and politics. And since ethics is the study and teaching of morally appropriate conduct, I have written that Confucianism is more accurately regarded as ethics.

To be regarded as a form of ethics is, in fact, an honor for Confucianism. The reason becomes clear in the next subsection (§4.4), where religion is compared with science, logic,

and ethics. Science, logic, and ethics possess no inherently undesirable features, whereas religion has many.

Of course, it is not easy to practice the teachings of Analects 2:3 or even 14:25 in today's world. For students, studying in order to gain admission to schools or secure employment — that is, to be recognized by examiners by outperforming competitors — is a pressing concern. Even university professors often study in order to retain their positions or secure research funding — that is, to gain recognition from superiors, colleagues, or society by surpassing their peers.

According to psychologists, people are most pleased when they are recognized by others. Likely, they are equally pleased when they surpass others. We can observe this not only through psychological experiments, but also in our daily interactions and social observations. Ultimately, as explained in §2.1 (**Systems with Life and Survival Instinct**), this tendency likely stems from the biological imperative of sexually reproducing organisms — including humans — to defeat same-sex rivals and gain recognition and acceptance from the opposite sex in order to leave offspring. In other words, the joy of being recognized by others is rooted in the biological instinct for survival.

To sublimate such deeply rooted tendencies, we might adopt the following perspective: Let us humbly and critically consider whether we and others are truly worthy of recognition — both of being recognized and of recognizing. Suppose that we are indeed worthy of recognition, and that we are recognized by someone else who is also worthy of recognition. Would that bring us joy as scholars? I think not. For to be recognized by another means only that we remain within the bounds of that person's understanding — we have not yet reached a higher summit beyond their view.

To use a metaphor: The height of Mount Fuji is evident even from its base, but the appearance of its summit cannot be discerned even from the peaks of neighboring mountains. We should aim to become such a lofty mountain. Such a goal may transform the joy of being recognized or surpassing others into the joy of self-realization. For Mount Fuji, the surrounding mountains and the lowlands — even if not submerged in clouds — are as good as nonexistent. This, I believe, is the true spirit of Analects 14:25.

If we restate the standard interpretation of Analects 14:25 in more logical terms, it becomes: "To be morally good, study not to gain recognition from others, but to cultivate yourself." Although the meaning of "to be morally good" is not entirely clear, this statement asserts that the act of studying for self-cultivation — rather than for recognition — is causally connected with being morally good.

Similarly, Analects 2:3 asserts that the act of guiding the people with virtue and propriety — rather than with laws and punishments — is causally connected with being morally good.

Generalizing from these examples, we may say that ethics is the study and teaching of actions that are causally connected with being morally good. Such actions are what I referred to earlier as "morally appropriate conduct."

However, the meaning of "morally good" is open to various interpretations. Accordingly, there are many kinds of ethics — such as social ethics and political ethics. In fact, the meaning of